

Marketplace category segmentation using revenue and sales volume.

Project Overview

The objective of this project was to perform a Price Architecture Analysis to understand pricing patterns across Amazon product categories.

Using category and sub-category level data, the analysis compared Average Price and Median Price across multiple segments to study price distribution and market structure. This helped identify how prices are spread within categories and whether pricing is influenced by premium outliers, discount-heavy products, or standardized market pricing.

The project focuses on examining price gaps, skewness, and consistency across categories to classify markets based on their pricing behavior, providing a clear view of how products are positioned within each category.

Data Set Summary

Dataset Context

The dataset, Amazon Sales Data 2023, contains detailed information on 551,585 products across 19 main categories and 112 sub-categories.

It includes 9 columns covering product details, pricing, customer ratings, and access information.

Key Features

- Product details: Name, Main Category, Sub-category
- Pricing: Discount Price, Actual Price
- Ratings: Average Rating, Number of Ratings
- Media & access: Image URL, Product Link

This dataset supports large-scale marketplace analysis across multiple product categories.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Loaded the cleaned CSV file into a DataFrame and previewed the first 15 records to understand the data layout and contents.

	name	main_category	sub_category	image	link	ratings	no_of_ratings	discount_price	actual_price
0	Lloyd 1.5 Ton 3 Star Inverter Split AC (5 In 1...	appliances	Air Conditioners	amazon.com/images/I/31UISB90sY...	https://www.amazon.in/Lloyd-Inverter-Convertible...	4.2	2255.0	32999.0	58990.0
1	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (C...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Convertible-Anti-Viru...	4.2	2948.0	46490.0	75990.0
2	LG 1 Ton 4 Star AI Dual Inverter Split AC (Copp...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Inverter-Convertible...	4.2	1206.0	34490.0	61990.0
3	LG 1.5 Ton 3 Star AI DUAL Inverter Split AC (C...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Convertible-Anti-Viru...	4.0	69.0	37990.0	68990.0
4	Carrier 1.5 Ton 3 Star Inverter Split AC (Copp...	appliances	Air Conditioners	amazon.com/images/I/41lrtqXPW...	https://www.amazon.in/Carrier-Inverter-Split-C...	4.1	630.0	34490.0	67790.0

- **Standardized numeric fields** by removing currency symbols, commas, and text to ensure consistent numerical formatting.
- **Converted relevant columns to numeric data types** to make them suitable for calculations and analysis.
- **Removed duplicate products** using a defined business key to avoid double counting in analysis.
- **Renamed columns for consistency and clarity**, improving readability and downstream usability.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions.

Due to the length and complexity of the queries, the detailed SQL code is omitted, and a set of reference slides is shared to illustrate the structure and contents of the analysis, without implying a complete or sequential narrative.

Why Price Architecture Matters

Customer Perspective Insights

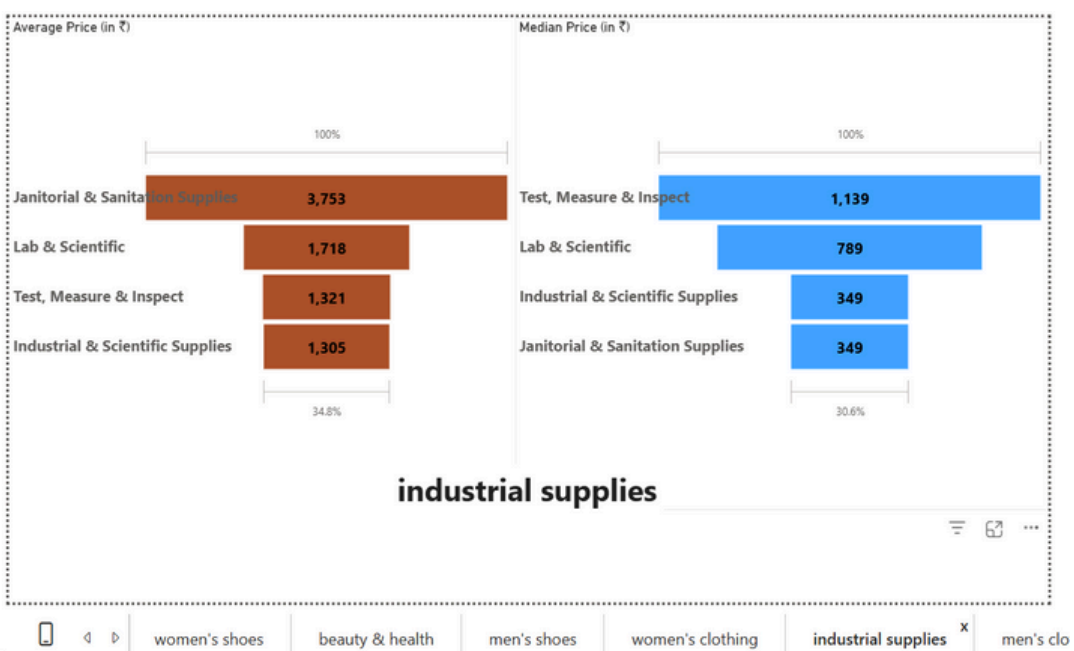
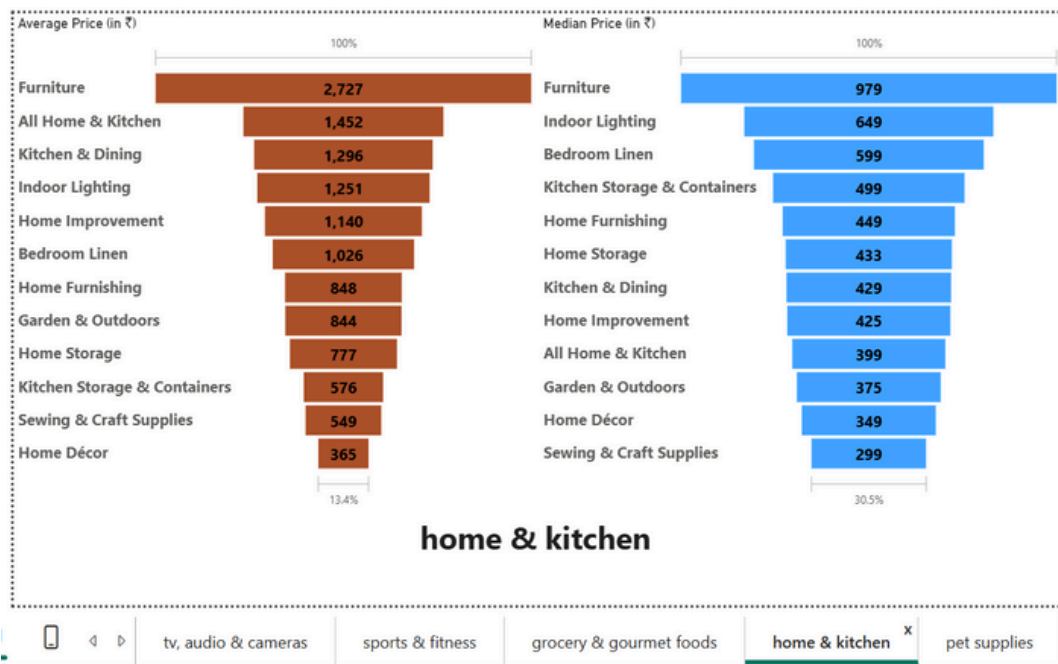
- * Reveals customer willingness to pay across different categories
- * Identifies price sensitivity and value perception in each segment
- * Highlights demand patterns and purchasing behavior trends

Competitive Market Positioning

- * Evaluates pricing position relative to competitors and market expectations
- * Identifies pricing gaps and opportunities within categories
- * Assesses whether products are positioned as budget, standard, or premium offerings

Strategic Business Decisions

- * Fine-tunes pricing strategies to maximize revenue and maintain market relevance
- * Enables data-driven decisions for product positioning, marketing campaigns, and discount strategies
- * Prevents decision-making based on "gut feel" by rooting pricing in real-world evidence



Executive Summary

Key Understanding from the Report :

Market is divided into 4 Parts

1.High revenue - Low Volume Categories

This category represents a strategic focus area for two key reasons: first, the market is not saturated, allowing easier entry; second, the high selling price points indicate substantial profit margin potential.

Findings To dominate in this Market:

- 1.Focus resources on specialized expertise over mass marketing
- 2.Build strategic partnerships to expand reach and credibility
- 3.Prioritize customer retention through exceptional personalized service experiences

2.Low Revenue - High Volume Categories

This category presents unique challenges requiring strategic consideration for two key reasons: first, the market is highly saturated with numerous competitors demanding differentiation; second, the low price points necessitate achieving profitability primarily through sales volume and operational efficiency

Findings to dominate in this Market:

- 1.Implement aggressive cost reduction through bulk purchasing and efficient supply chains
- 2.Establish wide distribution networks ensuring maximum product accessibility and convenience
- 3.Leverage mass advertising channels like TV commercials and traditional media for broad reach
- 4.Develop loyalty programs and promotional strategies to drive repeat purchases at volume
- 5.Optimize inventory turnover rates to compensate for thin profit margins.

3.Low Revenue - Low Volume Categories

This category represents the most challenging market position for two critical reasons: first, both revenue per unit and sales volume are insufficient to achieve sustainable profitability; second, the market demonstrates either declining demand or represents an untapped/unaware segment requiring significant customer education and market development investments.

Findings to dominate in this Market:

- 1.Pivot strategy by identifying alternative use cases or repositioning product for entirely different market segments with higher awareness
- 2.Conduct intensive market education campaigns to create category awareness where demand is latent rather than actively declining
- 3.Implement rapid iteration cycles based on early adopter feedback to solve unique unmet problems before scaling investment

4.High Revenue - High Volume Categories

This category represents the most competitive and lucrative market position for two critical reasons: first, the market is highly saturated with numerous established players aggressively defending their market share; second, winning in this segment can create substantial revenue and profit impact, making it the primary battleground for industry leaders.

Findings to dominate in this Market:

- 1.Implement preemptive defense strategies by launching innovations and advanced features before competitors can react or gain momentum
- 2.Build sustainable competitive moats through proprietary technology, exclusive partnerships, or distribution networks that competitors cannot easily replicate
- 3.Execute flanking defense by identifying and fortifying vulnerable market segments where competitors might attack or smaller players might gain traction
- 4.Deploy counter-offensive tactics with aggressive marketing campaigns and competitive pricing responses immediately when

High Revenue - Low Volume

1. TV, Audio & Cameras

* Television [₹1.7 lakh crore - 3k products]

2. Stores

* Amazon Fashion [₹22 thousand crore - 7k products]

3. Toys and Baby Products

* Baby products [₹21.4 thousand crore - 3.2k products]

* Diapers [₹21 thousand crore - 3.2k products]

4. Home & Kitchen

* Kitchen and dining [₹5300 crore - 3.6k products]

* Home improvement [₹4400 crore - 3.1k products]

5. Beauty and Health

* Value Bazar [₹5102 crore - 200 products]

* Health and personal care [₹8400 crore - 3300 products]

6. Industrial Supplies

* Janitorial & sanitation supplies [₹3730 crore - 2.7k products]

* Industrial & scientific supplies [₹3080 crore - 1.9k products]

7. Car & Motorbike

* Car electronics [₹2770 crore - 3k products]

8. Grocery and Gourmet Foods

* All grocery and gourmet foods [₹2180 crore - 2.9k products]

9. Pet Supplies

* Dog supplies [₹2820 crore - 2.9k products]

* All pet supplies [₹1722 crore - 1944 products]

Low Revenue - High Volume

1. Accessories

* Gold & diamond jewellery [₹2500 crore - 57k products]

2. Home & Kitchen

* Home furnishing [₹3400 crore - 4.2k products]

* Furniture [₹4000 crore - 4k products]

3. Men's Shoes

* Formal shoes [₹3800 crore - 58k products]

4. Women's Clothing

* Western wear [₹5100 crore - 58k products]

* Ethnic wear [₹7700 crore - 57k products]

* Lingerie & nightwear [₹6900 crore - 57k products]

5. Sports & Fitness

* Camping & hiking [₹1200 crore - 2400 products]

* Strength training [₹1500 crore - 3.3k products]

* Yoga [₹484 crore - 3.3k products]

* Running [₹840 crore - 2.7k products]

* Fitness accessories [₹880 crore - 3.9k products]

6. Men's Clothing

* Jeans [₹1800 crore - 28k products]

* T-shirts & polos [₹3430 crore - 57k products]

* Shirts [₹2930 crore - 58k products]

7. Industrial Supplies

* Test measure & inspect [₹1610 crore - 4.3k products]

8. Car & Motorbike

* Motorbike, accessories & parts [₹650 crore - 3.7k products]

* Car parts [₹290 crore - 3.7k products]

9. Bags & Luggage

* Travel accessories [₹1530 crore - 5.5k products]

* Rucksacks [₹1230 crore - 6.6k products]

* Suitcase & trolley bags [₹810 crore - 3.5k products]

* Travel duffles [₹242 crore - 3.5k products]

10. Kid's Fashion

* Kid's clothing [₹600 crore - 6.2k products]

* Baby fashion [₹470 crore - 6.6k products]

* Kid's watches [₹420 crore - 6k products]

* Kid's shoes [₹340 crore - 6.3k products]

11. Grocery and Gourmet Foods

* Snacks food [₹550 crore - 3.2k products]

12. Women's Shoes

* Fashion sandals [₹120 crore - 6.6k products]

* Ballerinas [₹140 crore - 4.2k products]

Low Revenue - Low Volume

1. Accessories

Sunglasses [₹330 crore - 4k products]

Beauty and Health

Luxury beauty [₹2200 crore - 2.6k products]

Sports & Fitness

Cardio Equipment [₹1000 crore - 700 products]

High Revenue - High Volume

1. Accessories

* Bags & luggage [₹28 thousand crore - 57k products]

2. Men's Shoes

* Sports shoes [₹24.4 thousand crore - 58k products]

* Casual shoes [₹21.2 thousand crore - 57k products]

3. Beauty and Health

* Beauty and grooming [₹5900 crore - 5.8k products]

4. Women's Clothing

* Clothing [₹13 thousand crore - 57k products]

5. Men's Clothing

* Innerwear [₹8450 crore - 57k products]

6. Car & Motorbike

* All car & motorbike products [₹2750 crore - 3.8k products]

7. Bags & Luggage

* Backpacks [₹2680 crore - 6.2k products]

* Wallets [₹1780 crore - 6k products]

8. Kids Fashion

* Kids fashion [₹2970 crore - 8.2k products]

* School bags [₹2180 crore - 7.1k products]

9. Women's Shoes

* Shoes [₹1490 crore - 5.6k products]